

Problem Set 4

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Data and Sample Selection

The data come from the PSID (Panel Study of Income Dynamics). We construct the hourly wage as labor income divided by annual hours worked. The following restrictions are applied to the sample:

- Male heads of household only.
- Age between 25 and 60 years.
- Hourly wage greater than \$7.

The dependent variable in all regressions is the log of the hourly wage. Key regressors include years of schooling (`educ`), age and age squared, and indicator variables for race/ethnicity (Black, Hispanic, and Other).

Likelihood Function

We estimate the following log-wage regression model:

$$\ln(w_{it}) = \beta_0 + \beta_1 \text{Educ}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Age}_{it}^2 + \beta_4 \text{Black}_{it} + \beta_5 \text{Hispanic}_{it} + \beta_6 \text{OtherRace}_{it} + \varepsilon_{it},$$

where $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$.

The likelihood function for parameters β and σ is:

$$L(\beta, \sigma \mid y, X) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - X_i\beta)^2}{2\sigma^2}\right).$$

Taking logs, the log-likelihood is:

$$\ell(\beta, \sigma \mid y, X) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - X_i\beta)^2.$$

The negative log-likelihood (minimized during estimation) is simply

$$-\ell(\beta, \sigma \mid y, X).$$

Results and Interpretation

Table 1 compares the OLS and MLE estimates of the education coefficient (β_1) across selected years.

Year	OLS $\hat{\beta}_1$	MLE $\hat{\beta}_1$
1971	0.067	0.067
1980	0.066	0.066
1990	0.096	0.096
2000	0.110	0.110

Table 1: Comparison of OLS and MLE estimates of the education coefficient.

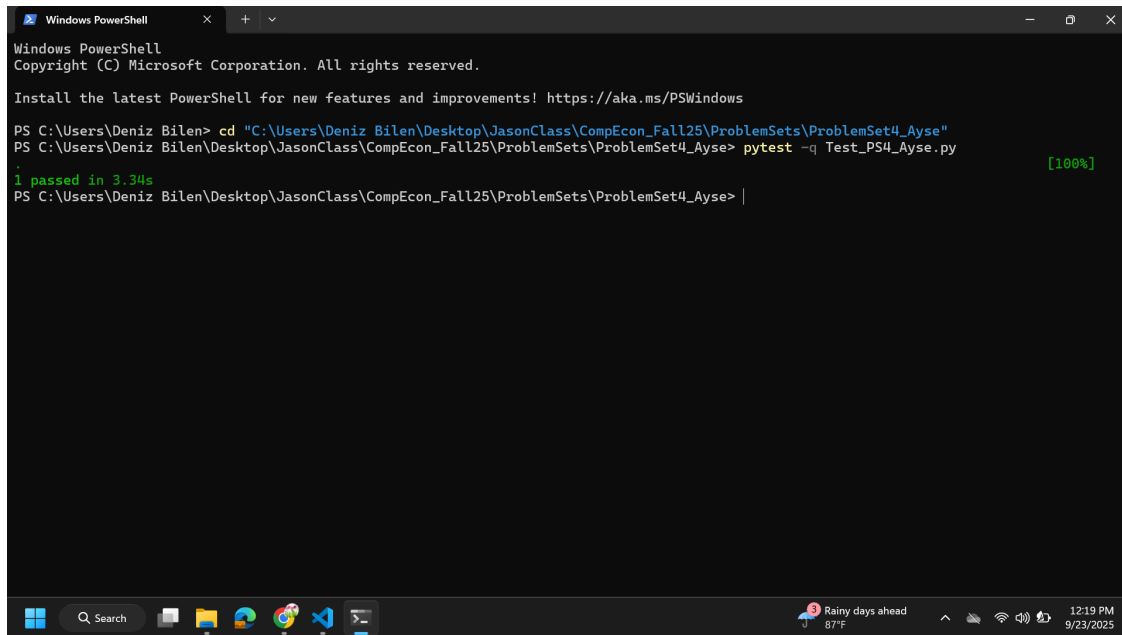
The coefficient on education (β_1) measures the approximate percentage increase in hourly wages associated with an additional year of schooling. The estimates show:

- In 1971, returns to schooling were about 6.7%.
- In 1980, they remained roughly constant at 6.6%.
- By 1990, they rose to about 9.6%.
- By 2000, they further increased to around 11.0%.

Overall, these results indicate that returns to education were stable during the 1970s and 1980s but increased substantially in the 1990s and 2000s, suggesting that education became increasingly rewarded in the labor market.

Unit Test

Figure 1 shows a screenshot of successful test results when running `pytest` in the terminal.



```
Windows PowerShell
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PS C:\Users\Deniz Bilen> cd "C:\Users\Deniz Bilen\Desktop\JasonClass\CompEcon_Fall25\ProblemSets\ProblemSet4_Ayse"
PS C:\Users\Deniz Bilen\Desktop\JasonClass\CompEcon_Fall25\ProblemSets\ProblemSet4_Ayse> pytest -q Test_PS4_Ayse.py [100%]
1 passed in 3.34s
PS C:\Users\Deniz Bilen\Desktop\JasonClass\CompEcon_Fall25\ProblemSets\ProblemSet4_Ayse> |
```

Figure 1: Successful `pytest` run for unit tests.